PROJECT REPORT

Modelling of binding of two 16-amino-acid biotinylated peptides to BSA

Peptides	A: CFAGTPSILKKNGGGS | B: KKAGTPSILMLAGGGS
Target	Bovine serum albumin (BSA), 583 aa, 66.5 kDa
Modifications	N-biotin (BTN, res 1), C-amide (NH2, res 16)
Structure prediction	Boltz-2.1 co-folding, 5 models per peptide
Binding affinity	PRODIGY (contact-based ΔG / Kd, 25 °C)
Reference project	CCB100725-YW01 (identical protocol)
Date	2026-07-14

Executive summary

Peptide A (CFAGTPSILKKNGGGS): predicted ΔG ranges -7.3 to -11.1 kcal/mol across 5 models; best model = Model 4 (ΔG -11.1 kcal/mol, Kd 6.8e-09 M, 92 interfacial contacts).

Peptide B (KKAGTPSILMLAGGGS): predicted ΔG ranges -8.1 to -9.8 kcal/mol across 5 models; best model = Model 4 (ΔG -9.8 kcal/mol, Kd 6.2e-08 M, 75 interfacial contacts).

Full per-model tables, interface-residue rankings, consensus hotspots and structure figures follow. A cross-comparison with the original peptide is given at the end.

Peptide A — CFAGTPSILKKNGGGS

Construct modeled

Biotin–CFAGTPSILKKNGGGS–NH2 (16-mer, N-biotinylated, C-amidated) bound to bovine serum albumin (BSA, 66.5 kDa, 583 aa).

Relative to the original (CFAGTPSILMLAGGGS): M10K, L11K, A12N — three central substitutions introducing two lysine residues (net +2 charge) into the C-terminal hydrophobic segment.

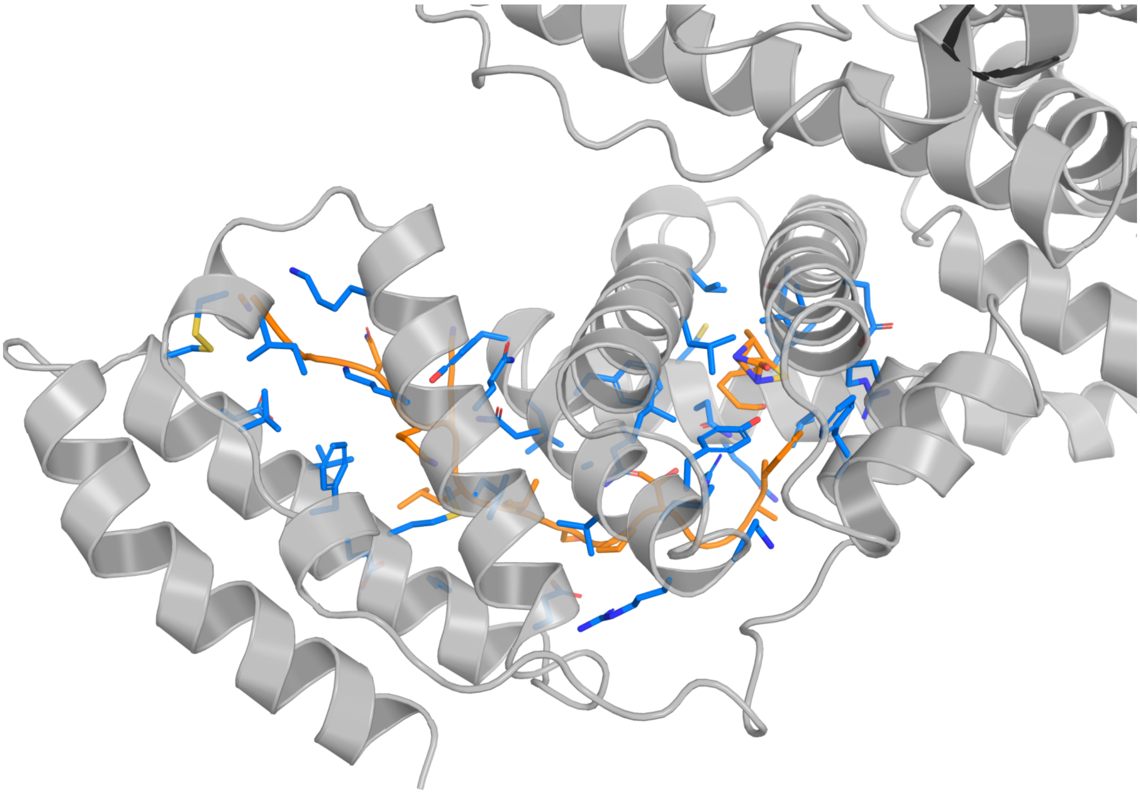
Method

Structure: Boltz-2.1 co-folding of the BSA–peptide complex (5 sampled models), biotin represented by CCD ligand BTN at residue 1 and C-terminal amidation by CCD NH2 at residue 16. Binding affinity / interface contacts: PRODIGY (contact-based predictor, 5.5 Å cutoff, 25 °C). Identical protocol to project CCB100725-YW01.

1) Model ranking and interface metrics

Model	ΔG (kcal/mol)	Kd (M @25°C)	Interfacial contacts	Hydrophobic contacts %	Apolar–apolar %
Model 1	-7.3	4.2e-06	55	70.9	21.8
Model 2	-8.6	4.9e-07	71	87.3	39.4
Model 3	-8.2	9.5e-07	56	80.4	35.7
Model 4	**-11.1**	**6.8e-09**	**92**	**84.8**	**31.5**
Model 5	-9.6	9.6e-08	73	86.3	38.4

Hydrophobic contacts % = (charged–apolar + apolar–polar + apolar–apolar)/total. Best predicted binder (lowest ΔG) highlighted.



Ray-traced PyMOL cartoon of the predicted BSA–peptide complex, best model (Model 4, ΔG -11.1 kcal/mol, Kd 6.8e-09 M). BSA (chain A) in gray cartoon (15% transparency); peptide (chain B) in orange cartoon + sticks; BSA residues within 5 Å of the peptide shown as marine-blue sticks.

2) Core interface residues (top-20 per model)

Columns: BSA residue (chain A), number of contacts to the peptide, and which peptide residues it touches.

Model 1 (ΔG -7.3 kcal/mol, 55 contacts)

Rank	BSA residue	Contacts	Peptide residues contacted
1	THR231	4	THR5, PRO6, SER7, ILE8
2	LYS232	4	PRO6, ILE8, LEU9, LYS10
3	THR235	4	THR5, PRO6, SER7, ILE8
4	TYR262	4	LEU9, LYS10, LYS11, ASN12
5	ASN266	4	LYS10, LYS11, ASN12, GLY13
6	ASP323	4	PHE2, ALA3, GLY4, THR5
7	ARG208	3	PHE2, ALA3, GLY4
8	LYS211	3	ALA3, GLY4, THR5
9	VAL228	3	ILE8, LEU9, LYS10
10	ALA212	2	PHE2, ALA3
11	VAL215	2	ALA3, THR5
12	GLU229	2	LYS10, ASN12
13	ASP265	2	ASN12, GLY13
14	THR269	2	LYS10, ASN12
15	GLU226	1	LYS10
16	VAL230	1	THR5
17	VAL234	1	THR5
18	ALA324	1	PRO6
19	LEU326	1	PHE2
20	GLY327	1	ALA3

Model 2 (ΔG -8.6 kcal/mol, 71 contacts)

Rank	BSA residue	Contacts	Peptide residues contacted
1	ALA324	5	PHE2, ALA3, GLY4, THR5, ILE8
2	GLU332	5	LYS11, ASN12, GLY13, GLY14, GLY15
3	GLU226	4	LYS10, LYS11, ASN12, GLY13
4	VAL228	4	PRO6, ILE8, LEU9, LYS10
5	THR231	4	GLY4, THR5, PRO6, ILE8
6	LYS211	3	PHE2, ALA3, GLY4
7	PHE227	3	GLY4, ILE8, LYS11
8	THR235	3	GLY4, THR5, PRO6
9	LEU301	3	GLY13, GLY14, GLY15
10	ASP307	3	LYS11, GLY14, GLY15
11	ASP323	3	PHE2, ALA3, GLY4
12	GLY327	3	PHE2, ALA3, GLY4
13	ARG335	3	GLY13, GLY14, GLY15
14	ARG208	2	PHE2, ALA3

15	ALA212	2	PHE2, ALA3
16	PRO302	2	GLY14, GLY15
17	SER328	2	PHE2, LYS11
18	TYR331	2	LYS11, GLY13
19	ARG336	2	GLY14, GLY15
20	VAL215	1	ALA3

Model 3 (ΔG -8.2 kcal/mol, 56 contacts)

Rank	BSA residue	Contacts	Peptide residues contacted
1	ALA324	5	THR5, PRO6, ILE8, GLY14, GLY15
2	PHE227	4	THR5, PRO6, ILE8, GLY15
3	THR231	4	THR5, PRO6, SER7, ILE8
4	ASP323	4	PHE2, ALA3, GLY4, THR5
5	LYS211	3	PHE2, ALA3, GLY4
6	VAL228	3	SER7, ILE8, LEU9
7	ASP307	3	LYS11, ASN12, GLY13
8	PHE325	3	GLY13, GLY14, GLY15
9	SER328	3	LYS11, GLY14, GLY15
10	ARG208	2	PHE2, ALA3
11	PRO302	2	LYS11, ASN12
12	ASN317	2	GLY13, GLY14
13	ALA321	2	GLY14, GLY15
14	GLY327	2	PHE2, THR5
15	GLU332	2	LYS11, ASN12
16	ALA209	1	PHE2
17	ALA212	1	PHE2
18	VAL215	1	THR5
19	GLU226	1	LYS11
20	THR235	1	THR5

Model 4 (ΔG -11.1 kcal/mol, 92 contacts)

Rank	BSA residue	Contacts	Peptide residues contacted
1	GLN393	5	GLY4, THR5, PRO6, SER7, ILE8
2	LEU397	5	PRO6, SER7, ILE8, LEU9, LYS10
3	ARG409	5	PHE2, ALA3, GLY4, THR5, PRO6
4	LYS524	5	LYS11, ASN12, GLY13, GLY14, GLY15
5	GLY401	4	ILE8, LEU9, LYS10, LYS11
6	ASN404	4	ILE8, LEU9, LYS10, LYS11
7	GLU540	4	THR5, PRO6, SER7, ILE8
8	MET547	4	ILE8, LEU9, LYS10, LYS11
9	VAL551	4	LYS10, LYS11, ASN12, GLY13

10	ASP555	4	ASN12, GLY13, GLY14, GLY15
11	GLN389	3	PHE2, ALA3, GLY4
12	ASN390	3	PHE2, ALA3, THR5
13	TYR400	3	LYS10, LYS11, ASN12
14	ALA405	3	THR5, PRO6, ILE8
15	LYS520	3	GLY13, GLY14, GLY15
16	VAL554	3	GLY13, GLY14, GLY15
17	LEU386	2	PHE2, ALA3
18	LEU406	2	PHE2, THR5
19	VAL408	2	PRO6, ILE8
20	ILE512	2	GLY14, GLY15

Model 5 (ΔG -9.6 kcal/mol, 73 contacts)

Rank	BSA residue	Contacts	Peptide residues contacted
1	LEU397	5	THR5, PRO6, SER7, ILE8, LYS10
2	ARG409	5	PHE2, ALA3, GLY4, THR5, PRO6
3	GLN393	4	THR5, PRO6, SER7, ILE8
4	ALA405	4	THR5, PRO6, ILE8, LEU9
5	VAL551	4	ASN12, GLY13, GLY14, GLY15
6	GLN389	3	ALA3, GLY4, THR5
7	ASN390	3	PHE2, ALA3, THR5
8	GLY401	3	ILE8, LEU9, LYS10
9	ASN404	3	ILE8, LEU9, GLY13
10	VAL408	3	PRO6, ILE8, LEU9
11	GLU540	3	PRO6, SER7, ILE8
12	LYS544	3	ILE8, LEU9, LYS11
13	LEU386	2	PHE2, ALA3
14	LYS396	2	SER7, LYS10
15	TYR400	2	GLY13, GLY14
16	LEU406	2	PHE2, THR5
17	LYS524	2	GLY14, GLY15
18	LEU543	2	ILE8, LEU9
19	ASP555	2	ASN12, GLY15
20	PHE402	1	THR5

3) BSA residues recurring across models

Seen in all/≥4 of 5 models (strong consensus): —

Seen in 3 of 5 models: ARG208, LYS211, ALA212, VAL215, GLU226, VAL228, THR231, THR235, ASP323, ALA324, LEU326, GLY327, LEU330, ALA349, GLU353

Seen in 2 of 5 models: PHE227, LYS232, VAL234, LEU301, PRO302, ASP307, LYS322, SER328, GLU332, ARG335, LEU346, LYS350, LEU386, GLN389, ASN390, GLN393, LYS396, LEU397, TYR400, GLY401, PHE402, ASN404, ALA405, LEU406, VAL408, ARG409, TYR410, ARG412, LYS413, LEU456, ARG484, PHE487, SER488,

LYS524, LEU528, GLU540, LEU543, LYS544, MET547, VAL551, VAL554, ASP555

Interface hotspot regions on BSA (residue-number bins, residues seen in ≥2 models):

- 180–209: ARG208
- 210–239: LYS211, ALA212, VAL215, GLU226, PHE227, VAL228, THR231, LYS232, VAL234, THR235
- 300–329: LEU301, PRO302, ASP307, LYS322, ASP323, ALA324, LEU326, GLY327, SER328
- 330–359: LEU330, GLU332, ARG335, LEU346, ALA349, LYS350, GLU353
- 360–389: LEU386, GLN389
- 390–419: ASN390, GLN393, LYS396, LEU397, TYR400, GLY401, PHE402, ASN404, ALA405, LEU406, VAL408, ARG409, TYR410, ARG412, LYS413
- 450–479: LEU456
- 480–509: ARG484, PHE487, SER488
- 510–539: LYS524, LEU528
- 540–569: GLU540, LEU543, LYS544, MET547, VAL551, VAL554, ASP555

4) Which peptide residues contact BSA most

Top peptide residues by interface contact count within each model.

Model	#1	#2	#3	#4	#5	#6
Model 1	PHE2 (9)	ALA3 (7)	THR5 (7)	LYS10 (7)	ASN12 (5)	PRO6 (4)
Model 2	PHE2 (14)	ALA3 (8)	GLY15 (8)	GLY4 (7)	LYS11 (7)	GLY14 (6)
Model 3	PHE2 (11)	THR5 (7)	LYS11 (7)	GLY14 (5)	GLY15 (5)	ILE8 (4)
Model 4	PHE2 (11)	ILE8 (10)	LYS11 (9)	LYS10 (8)	THR5 (7)	PRO6 (7)
Model 5	PHE2 (12)	ILE8 (11)	THR5 (8)	LEU9 (8)	PRO6 (7)	ALA3 (4)

Peptide B — KKAGTPSILMLAGGGS

Construct modeled

Biotin–(K)KAGTPSILMLAGGGS–NH2 (16-mer, N-biotinylated, C-amidated) bound to bovine serum albumin (BSA, 66.5 kDa, 583 aa).

Relative to the original (CFAGTPSILMLAGGGS): C1K, F2K — two N-terminal substitutions replacing Cys/Phe with two lysines (net +2 charge). As in the original protocol the residue-1 position is represented by the biotin cap, so the modeled construct is Biotin–K–AGTPSILMLAGGG–NH2 (first Lys represented by biotin).

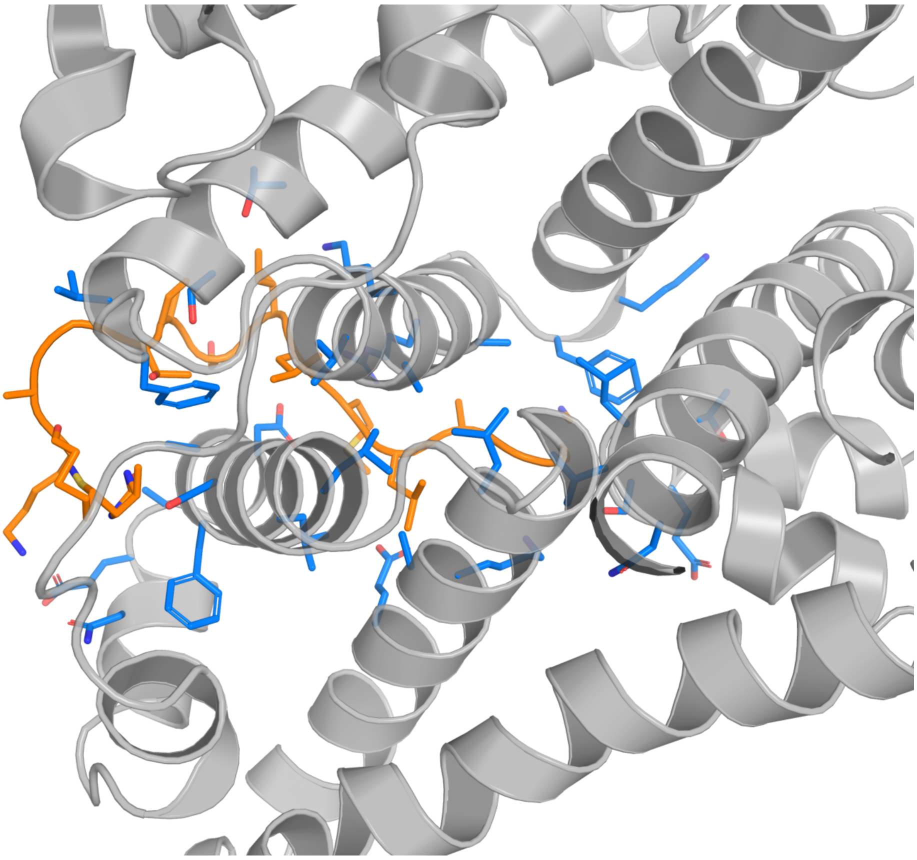
Method

Structure: Boltz-2.1 co-folding of the BSA–peptide complex (5 sampled models), biotin represented by CCD ligand BTN at residue 1 and C-terminal amidation by CCD NH2 at residue 16. Binding affinity / interface contacts: PRODIGY (contact-based predictor, 5.5 Å cutoff, 25 °C). Identical protocol to project CCB100725-YW01.

1) Model ranking and interface metrics

Model	ΔG (kcal/mol)	Kd (M @25°C)	Interfacial contacts	Hydrophobic contacts %	Apolar–apolar %
Model 1	-8.1	1.1e-06	64	90.6	39.1
Model 2	-9.2	1.9e-07	81	87.7	38.3
Model 3	-9.2	1.7e-07	80	86.2	40.0
Model 4	**-9.8**	**6.2e-08**	**75**	**92.0**	**41.3**
Model 5	-8.4	6.9e-07	78	88.5	43.6

Hydrophobic contacts % = (charged–apolar + apolar–polar + apolar–apolar)/total. Best predicted binder (lowest ΔG) highlighted.



Ray-traced PyMOL cartoon of the predicted BSA–peptide complex, best model (Model 4, ΔG -9.8 kcal/mol, Kd 6.2e-08 M). BSA (chain A) in gray cartoon (15% transparency); peptide (chain B) in orange cartoon + sticks; BSA residues within 5 Å of the peptide shown as marine-blue sticks.

2) Core interface residues (top-20 per model)

Columns: BSA residue (chain A), number of contacts to the peptide, and which peptide residues it touches.

Model 1 (ΔG -8.1 kcal/mol, 64 contacts)

Rank	BSA residue	Contacts	Peptide residues contacted
1	LYS136	5	THR5, SER7, ILE8, LEU9, MET10
2	LEU115	4	LEU9, MET10, LEU11, ALA12
3	GLU140	4	SER7, ILE8, MET10, ALA12
4	ARG144	4	ALA12, GLY13, GLY14, GLY15
5	HIS145	4	ALA12, GLY13, GLY14, GLY15
6	PRO117	3	LEU9, MET10, LEU11
7	LEU122	3	ILE8, LEU9, MET10
8	TYR160	3	LEU9, MET10, LEU11
9	ARG185	3	LEU11, ALA12, GLY13
10	LEU189	3	ALA12, GLY13, GLY14
11	GLN33	2	LYS2, ALA3
12	PRO35	2	LYS2, THR5
13	LYS116	2	LEU9, MET10
14	GLU125	2	SER7, ILE8
15	PHE133	2	ILE8, MET10
16	TYR137	2	MET10, LEU11
17	ILE141	2	LEU11, ALA12
18	GLU424	2	GLY14, GLY15
19	ARG458	2	GLY14, GLY15
20	ASP37	1	THR5

Model 2 (ΔG -9.2 kcal/mol, 81 contacts)

Rank	BSA residue	Contacts	Peptide residues contacted
1	LEU115	6	SER7, ILE8, LEU9, MET10, LEU11, ALA12
2	GLU140	5	THR5, SER7, ILE8, LEU9, ALA12
3	ARG185	5	MET10, LEU11, ALA12, GLY13, GLY15
4	PRO35	4	LYS2, ALA3, GLY4, THR5
5	LYS116	4	ILE8, LEU9, MET10, LEU11
6	PRO117	4	ILE8, LEU9, MET10, LEU11
7	LEU122	4	SER7, ILE8, LEU9, MET10
8	LYS136	4	THR5, PRO6, SER7, LEU9
9	ARG144	4	ILE8, ALA12, GLY13, GLY14
10	GLU125	3	SER7, ILE8, LEU9
11	HIS145	3	ALA12, GLY13, GLY14
12	LEU189	3	LEU11, ALA12, GLY13
13	PHE36	2	GLY4, THR5
14	GLU38	2	LYS2, ALA3

15	LYS114	2	ILE8, LEU11
16	ASP118	2	ILE8, LEU9
17	TYR137	2	LEU9, MET10
18	ILE141	2	MET10, ALA12
19	TYR160	2	LEU9, MET10
20	ILE181	2	LEU9, MET10

Model 3 (ΔG -9.2 kcal/mol, 80 contacts)

Rank	BSA residue	Contacts	Peptide residues contacted
1	ASP323	6	THR5, PRO6, SER7, ILE8, LEU9, MET10
2	ALA324	6	LYS2, ALA3, GLY4, THR5, PRO6, LEU9
3	ARG208	5	PRO6, SER7, ILE8, LEU9, MET10
4	LYS350	5	LEU11, ALA12, GLY13, GLY14, GLY15
5	LYS211	4	THR5, SER7, ILE8, LEU9
6	ALA212	4	ILE8, LEU9, MET10, LEU11
7	THR231	4	ALA3, GLY4, THR5, ILE8
8	SER328	4	LYS2, ALA3, GLY4, LEU9
9	GLU353	4	MET10, LEU11, GLY14, GLY15
10	PHE227	3	ALA3, GLY4, ILE8
11	GLU478	3	ALA12, GLY13, GLY14
12	PHE205	2	ALA12, GLY13
13	ALA209	2	MET10, ALA12
14	GLY327	2	ILE8, LEU9
15	LEU330	2	LEU9, LEU11
16	SER479	2	ALA12, GLY13
17	LEU480	2	ALA12, GLY13
18	ASN482	2	ALA12, GLY13
19	VAL215	1	ILE8
20	VAL228	1	ALA3

Model 4 (ΔG -9.8 kcal/mol, 75 contacts)

Rank	BSA residue	Contacts	Peptide residues contacted
1	ASP323	7	THR5, PRO6, SER7, ILE8, LEU9, MET10, LEU11
2	ARG208	6	SER7, ILE8, LEU9, MET10, LEU11, ALA12
3	PHE205	4	ALA12, GLY13, GLY14, GLY15
4	ALA212	4	LEU9, MET10, LEU11, ALA12
5	PHE227	4	ALA3, GLY4, THR5, LEU9
6	THR231	4	GLY4, THR5, PRO6, LEU9
7	ALA324	4	LYS2, THR5, PRO6, LEU9
8	ALA209	3	MET10, ALA12, GLY13
9	LYS211	3	ILE8, LEU9, MET10

10	LYS350	3	LEU11, GLY13, GLY14
11	GLU478	3	GLY13, GLY14, GLY15
12	SER479	3	GLY13, GLY14, GLY15
13	LEU480	3	ALA12, GLY13, GLY14
14	VAL481	3	ALA12, GLY13, GLY14
15	VAL228	2	ALA3, GLY4
16	THR235	2	PRO6, ILE8
17	GLU353	2	MET10, LEU11
18	THR477	2	GLY14, GLY15
19	LYS204	1	GLY15
20	VAL215	1	LEU9

Model 5 (ΔG -8.4 kcal/mol, 78 contacts)

Rank	BSA residue	Contacts	Peptide residues contacted
1	LEU115	5	LEU11, ALA12, GLY13, GLY14, GLY15
2	PRO420	5	THR5, PRO6, SER7, ILE8, LEU9
3	ARG185	4	LEU11, ALA12, GLY13, GLY14
4	LEU505	4	LYS2, ALA3, GLY4, THR5
5	LYS114	3	SER7, ILE8, LEU9
6	PRO117	3	GLY13, GLY14, GLY15
7	ILE141	3	ALA12, GLY13, GLY14
8	ARG144	3	MET10, LEU11, ALA12
9	HIS145	3	MET10, LEU11, ALA12
10	TYR160	3	GLY13, GLY14, GLY15
11	LEU189	3	MET10, LEU11, ALA12
12	LYS504	3	LYS2, ALA3, GLY4
13	ASP111	2	ILE8, MET10
14	LEU112	2	ILE8, MET10
15	LEU122	2	GLY14, GLY15
16	LYS136	2	GLY14, GLY15
17	TYR137	2	GLY13, GLY14
18	GLU140	2	GLY14, GLY15
19	ILE181	2	GLY13, GLY14
20	VAL423	2	SER7, LEU9

3) BSA residues recurring across models

Seen in all/≥4 of 5 models (strong consensus): —

Seen in 3 of 5 models: LYS114, LEU115, LYS116, PRO117, LEU122, GLU125, LYS136, TYR137, GLU140, ILE141, ARG144, HIS145, TYR160, ILE181, ARG185, LEU189, GLU424, ARG458

Seen in 2 of 5 models: PRO35, ASP37, ASP108, SER109, PRO110, ASP118, PHE133, VAL188, PHE205, ARG208, ALA209, LYS211, ALA212, VAL215, PHE227, VAL228, THR231, VAL234, THR235, ASN317, GLU320, ALA321, ASP323, ALA324, LEU326, GLY327, SER328, LEU330, LEU346, ALA349, LYS350, GLU353, GLU478, SER479,

LEU480, VAL481, ASN482

Interface hotspot regions on BSA (residue-number bins, residues seen in ≥2 models):

- 30–59: PRO35, ASP37
- 90–119: ASP108, SER109, PRO110, LYS114, LEU115, LYS116, PRO117, ASP118
- 120–149: LEU122, GLU125, PHE133, LYS136, TYR137, GLU140, ILE141, ARG144, HIS145
- 150–179: TYR160
- 180–209: ILE181, ARG185, VAL188, LEU189, PHE205, ARG208, ALA209
- 210–239: LYS211, ALA212, VAL215, PHE227, VAL228, THR231, VAL234, THR235
- 300–329: ASN317, GLU320, ALA321, ASP323, ALA324, LEU326, GLY327, SER328
- 330–359: LEU330, LEU346, ALA349, LYS350, GLU353
- 420–449: GLU424
- 450–479: ARG458, GLU478, SER479
- 480–509: LEU480, VAL481, ASN482

4) Which peptide residues contact BSA most

Top peptide residues by interface contact count within each model.

Model	#1	#2	#3	#4	#5	#6
Model 1	MET10 (9)	LEU9 (8)	LEU11 (8)	ILE8 (7)	ALA12 (7)	GLY15 (7)
Model 2	LEU9 (13)	MET10 (11)	ILE8 (10)	THR5 (7)	ALA12 (7)	LEU11 (6)
Model 3	LEU9 (11)	ILE8 (10)	ALA12 (8)	LYS2 (6)	LEU11 (6)	GLY13 (6)
Model 4	LEU9 (11)	LEU11 (9)	GLY13 (8)	GLY14 (7)	MET10 (6)	ALA12 (6)
Model 5	MET10 (10)	GLY14 (10)	GLY15 (8)	LYS2 (7)	ALA12 (7)	GLY13 (7)

5) Cross-comparison: new peptides vs. original

Peptide	Sequence	Best ΔG (kcal/mol)	Best Kd (M)	Mean ΔG	Max contacts
Peptide A	CFAGTPSILKKNGGGS	-11.1	6.8e-09	-9.0	92
Peptide B	KKAGTPSILMLAGGGS	-9.8	6.2e-08	-8.9	81
Original	CFAGTPSILMLAGGGS	-12.4	8.6e-10	-10.0	102

Predicted affinity: all three peptides are predicted to bind BSA with sub-micromolar affinity in their best models. The original peptide remained the strongest single model (ΔG -12.4 kcal/mol, Kd 8.6e-10 M). Among the new peptides, Peptide A reaches the most favourable single model (Model 4, ΔG -11.1 kcal/mol, Kd 6.8e-9 M, 92 contacts) and Peptide B is more tightly clustered and reproducible (ΔG -8.1 to -9.8 kcal/mol across all 5 models).

Binding-site behaviour — Peptide A (M10K/L11K/A12N): notably, its best-scoring Model 4 re-docks into the SAME canonical BSA pocket used by the original peptide (GLN393, LEU397, GLY401, ASN404, ARG409, GLU540, MET547, VAL551 — the 390-410 / 540-548 patch), whereas its other four models occupy an alternative groove around residues 208-353 (ARG208/LYS211/ALA212, ASP323/ALA324, GLU353). The added lysines therefore make the alternative charged groove competitive but do not abolish the original hydrophobic mode.

Binding-site behaviour — Peptide B (C1K/F2K): all five models converge on a distinct, more N-terminal/charged region of BSA around residues 114-189 (LYS114/LYS116, GLU125/GLU140, ARG144/HIS145, ARG185), recruiting Arg/Lys/Glu/His to complement the two new N-terminal lysines. This site is consistent and reproducible (higher inter-model agreement, protein-iptm up to 0.76).

Interpretation: the substitutions do not remove BSA binding; they re-route or diversify it. BSA is a promiscuous carrier with several hydrophobic and charged sub-sites, so the biotinylated peptides retain predicted sub-micromolar affinity even after inserting positive charge. All ΔG/Kd values are Boltz-2.1 (structure) + PRODIGY (affinity) predictions and should be confirmed experimentally — e.g. BLI or SPR with the biotinylated peptides captured on streptavidin sensors against BSA.

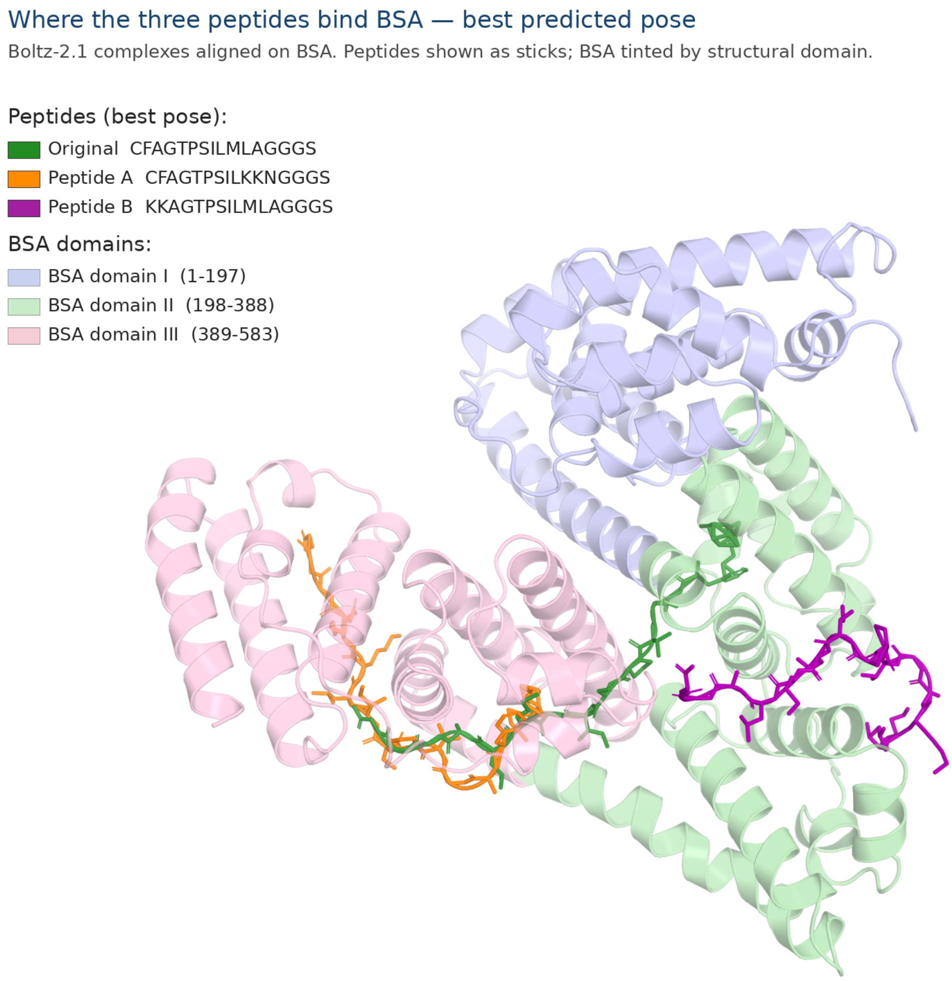
Method note: the biotin (BTN) and C-terminal amide (NH2) caps occupy peptide positions 1 and 16 respectively and are excluded from the PRODIGY contact analysis (which requires standard residues), exactly as in project CCB100725-YW01; the interface analysis therefore reflects peptide residues 2-15.

6) Global binding-site map on BSA

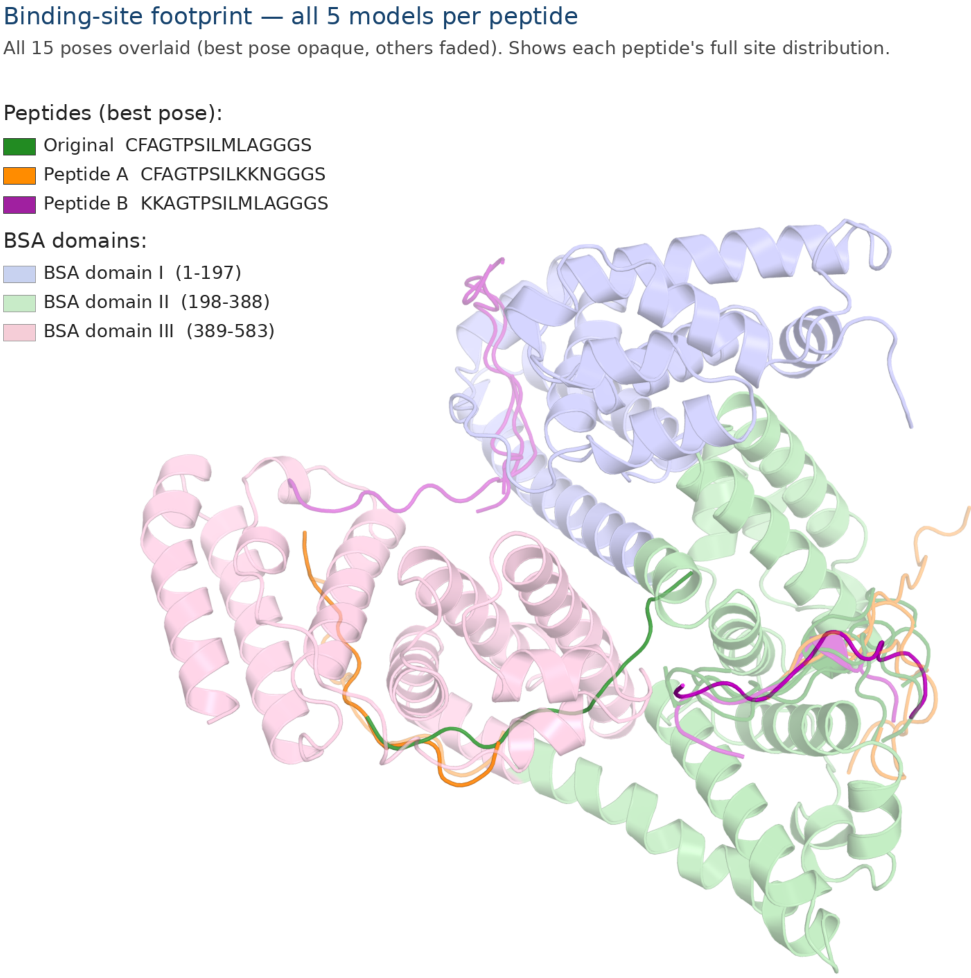
All Boltz-2.1 complexes were superimposed on the shared BSA chain and the peptides overlaid, so the three binding sites can be compared in one frame. BSA is coloured by its three structural domains: I (residues 1–197), II (198–388), III (389–583).

Peptide	Sequence	Best-pose site (ΔG)	Across 5 models
Original	CFAGTPSILMLAGGGS	IIIA (-14.4)	4/5 dom II, 1/5 dom III
Peptide A	CFAGTPSILKKNGGGS	IIIA (-11.1)	3/5 dom II, 2/5 dom III
Peptide B	KKAGTPSILMLAGGGS	IIA (-9.8)	3/5 dom I, 2/5 dom II

Original and **Peptide A** share the classic domain II/III hydrophobic pockets (Sudlow site I in subdomain IIA and site II in IIIA); Peptide A's strongest pose occupies the same IIIA patch as the original peptide. **Peptide B** is the outlier — its two N-terminal lysines let it uniquely engage **domain I** (subdomain IB, ~residues 114–189) in the majority of models. All three are somewhat promiscuous, as expected for BSA (a multi-site carrier protein).



Best predicted pose of each peptide on BSA (Original = green, Peptide A = orange, Peptide B = purple; BSA tinted by domain).

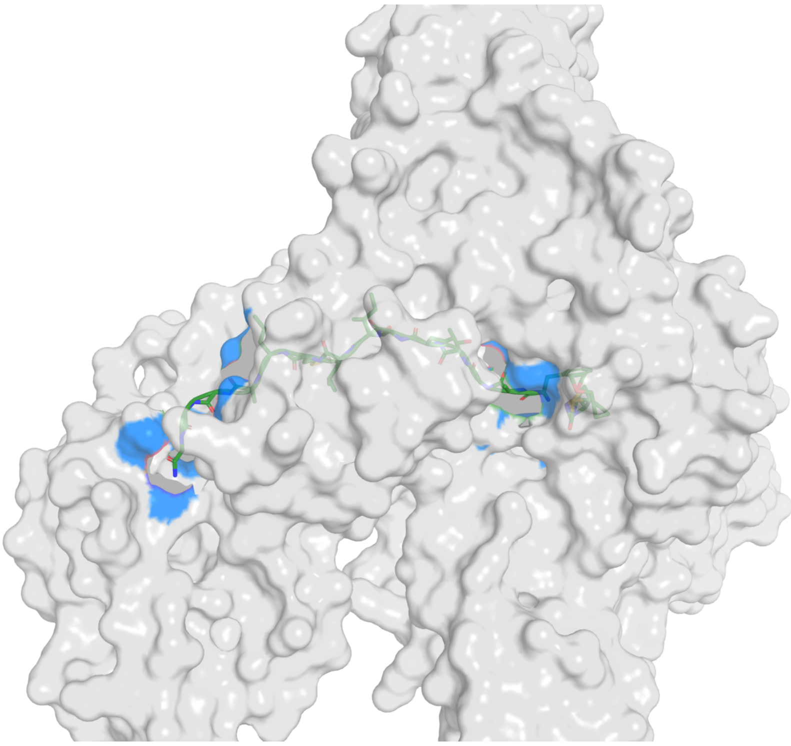


Binding-site footprint — all 5 models per peptide overlaid (best pose opaque, others faded). Peptide B's poses reaching into domain I (blue, upper-left) are unique to it.

BSA surface pockets (best pose)

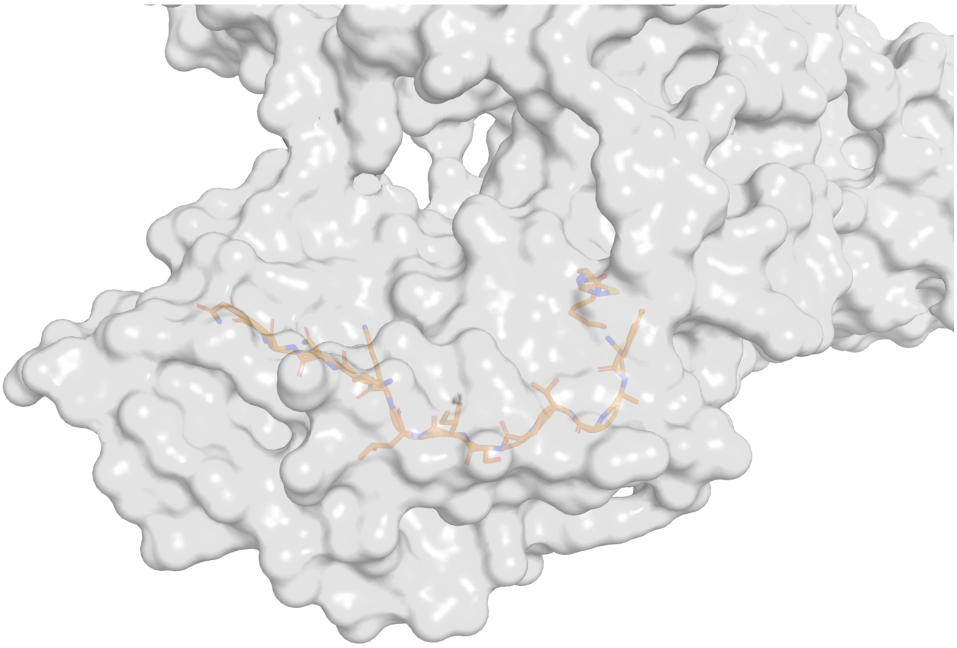
BSA drawn as a semi-transparent molecular surface with each peptide (sticks + cartoon) resting in its binding groove.

Original — CFAGTPSILMLAGGGS



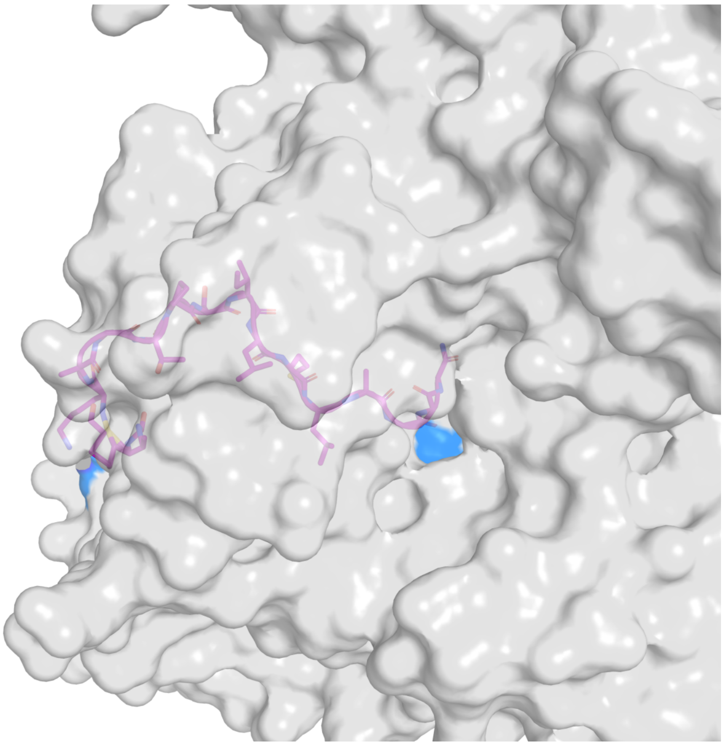
Peptide (sticks + cartoon) in its BSA surface groove; marine patches = surface directly contacting the peptide.

Peptide A — CFAGTPSILKKNGGGS



Peptide (sticks + cartoon) in its BSA surface groove; marine patches = surface directly contacting the peptide.

Peptide B — KKAGTPSILMLAGGGS



Peptide (sticks + cartoon) in its BSA surface groove; marine patches = surface directly contacting the peptide.